ES_APPM 421 Final Project Report

Modeling Inhomogeneous Broadening through Spin-Spin Interactions in Matrix Isolated Spin Ensembles

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I. INTRODUCTION

Many modern physics experiments use atoms as very precise sensors. All atoms have "discrete energy levels," meaning they require an *exact* amount of energy to change from one energy level (or state) to another. We can supply this energy using light with a very specific frequency and then measure how much light is absorbed. If we place the atoms in different environments, such as in a magnetic field, these well-known energy levels can shift slightly. By measuring how much the energy levels shift (by looking at how the absorption of light changes) we can learn about the environment. In this way, atoms can act as extremely sensitive sensors, and are used in some of the most precise measurements of magnetic fields and even in searches for new physics, such as possible effects of dark matter.

The sensitivity of these sensors is limited by something called *inhomogeneous broadening*. If we look at many atoms at once, ideally their discrete energy levels would all be spaced in exactly the same way. In practice this does not happen. Atoms can interact slightly differently with each other and with their surroundings, so each atom's energy levels shift by a slightly different amount. When we measure the signal from all the atoms together, these small differences add up and smear out the signal, making it broader instead of sharp. This broadening limits how precisely we can tell what the energy levels really are, and therefore limits our sensitivity.

A specific system where this inhomogeneous broadening is currently a limiting factor is called a *matrix-isolated spin ensemble*. In this system, a huge number of atoms (the *dopants*) are trapped, or frozen in place, inside a solid crystal matrix made from some inert material, typically a noble gas like neon. Using more atoms should, in principle, improve sensitivity, because having more atoms generally gives a stronger signal. However, in practice, any improvement we gain by increasing the number of dopant atoms has been largely lost because the inhomogeneous broadening is so large.

Within the matrix-isolated spin-ensemble community there is still debate about what is causing this broadening. Some researchers argue it is mainly due to structural defects in the crystal, such as cracks or local distortions in the solid matrix. Others argue that it comes from spin-spin interactions, where the dopant atoms interact magnetically and disturb each other. This research area is relatively small and young, with only a few groups working on it, and to my knowledge there has not been a published mathematical modeling approach that directly simulates how these spins might broaden the signal.

Therefore, in this project I focus on modeling one possible source of broadening: spin-spin interactions. I treat each dopant atom as a tiny bar magnet (a magnetic dipole) and use Monte Carlo methods to simulate many different possible arrangements of atoms inside a matrix-isolated spin ensemble. For each arrangement, I calculate how the neighboring atoms shift the energy levels of a central "test" atom and how this leads to a spread in the measured signal. My initial hypothesis was that these spin-spin interactions will cause larger broadening than some people currently expect and could explain the large inhomogeneous broadening seen in experiments. The rest of this report describes the model I use (Sec. II), how I analyze the simulated data (Sec. III), and how the results compare to experimental measurements (Sec. IV).

II. MATHEMATICAL MODEL

A. Atom/Spin Model

Despite the complex physics making up this physical system, we can get away with just some basic understanding in order to follow this model. If you wish to understand the underlying quantum physics in more detail, I suggest looking at Ref. [1]. Below, I present only the basics necessary for following this project.

In quantum physics, atoms have discrete energy levels, meaning that for an atom to "transition" or go from one energy level to another, requires a specific amount of energy, typically in the form of light. In this project, I will refer to these energy levels as *spin states*. Atoms can have many spin states, so to specify which ones we are

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talking about, we use two main integer variables:

- F, the first variable, formally called the "total atomic angular momentum."
- m_F, the second variable, formally called the "magnetic quantum number." Its value lies between $-F$ and $+F$.

For this model, F and m_F behave as input labels that are defined by the atom and spin state we are modeling. The detailed physical meaning behind them is not important for following the calculations here.

The notation we use to define a spin state is

$$|F, m_F\rangle\,, \tag{1}$$

which means the energy level that corresponds to the inputs F and m_F.

In this model, we will specifically look at the energy difference between two different spin states. To represent this we use

$$|F_1, m_{F,1}\rangle \rightarrow |F_2, m_{F,2}\rangle\,, \tag{2}$$

where the difference in energy between these two states is E_0. In practice, one uses light (like a laser) to provide this energy. The frequency ν_0 of light needed to achieve this energy difference is

$$\nu_0 = \frac{E_0}{h}, \tag{3}$$

where $h \approx 6.626 \times 10^{-34}$ J s is Planck's constant. Fig. 1 shows an energy diagram of how light with energy $h\nu_0$ produces the transition between two spin states.



FIG. 1. Schematic two-level diagram: light of frequency ν_0 drives a transition between two spin states separated by energy E_0.

Two additional variables, which do not define the spin states themselves but are necessary for differentiating different types of atoms and the strength of their interactions, are

- I, formally called the "nuclear angular momentum"
- J, formally called the "total electron angular momentum." In all the systems we model for this project $J = \frac{1}{2}$.

Again, in this model these are just inputs, determined by the choice of atom. Table I lists the relevant values of I, and the allowed F and m_F values, for the atoms and spin states we will consider in this project.

Element	Isotope	I	F	m_F
Sodium	^{23}Na	$\frac{3}{2}$	1	$-1, 0, 1$
			2	$-2, -1, 0, 1, 2$
Potassium	^{39}K	$\frac{3}{2}$	1	$-1, 0, 1$
			2	$-2, -1, 0, 1, 2$
Rubidium	^{85}Rb	$\frac{5}{2}$	2	$-2, -1, 0, 1, 2$
			3	$-3, -2, -1, 0, 1, 2, 3$
Rubidium	^{87}Rb	$\frac{3}{2}$	1	$-1, 0, 1$
			2	$-2, -1, 0, 1, 2$
Cesium	^{133}Cs	$\frac{7}{2}$	3	$-3, -2, -1, 0, 1, 2, 3$
			4	$-4, -3, -2, -1, 0, 1, 2, 3, 4$

TABLE I. I, F, and corresponding m_F values for atoms and spin states considered in this project.

B. Spin-Spin Interaction Model

When we place an atom with spin near other atoms with spin, we expect these spins to interact and shift the resulting energy of the spin states. Doing a full quantum-mechanical solution of N_{spins} interacting spins requires solving a matrix of size $2^{N_{\text{spins}}} \times 2^{N_{\text{spins}}}$ [2]. This quickly becomes computationally expensive for large systems, such as the matrix-isolated spin ensembles I wish to model.

However, the primary way these spins interact is through the magnetic fields they produce. Therefore, in my project I model each spin as a magnetic field source and calculate how the resulting magnetic field shifts the energy of a central test spin whose transition we are interested in. The equations needed to do this are summarized below.

To first order, the magnetic field from a spin looks like the field from a bar magnet. These are *dipole* magnets and create a magnetic field at location $\boldsymbol{r}$ of

$$\boldsymbol{B}(\boldsymbol{r}, \boldsymbol{m}) = \frac{\mu_0}{4\pi r^3}\big[3\hat{\boldsymbol{r}}(\boldsymbol{m}\cdot\hat{\boldsymbol{r}}) - \boldsymbol{m}\big]. \tag{4}$$

Here $\mu_0 \approx 4\pi \times 10^{-7}$ T m/A is the permeability of free space constant, and $\boldsymbol{m}$ is the *magnetic moment* of the dipole. For the spin states we are modeling,

$$\boldsymbol{m} = \mu_B g_F m_F \hat{\boldsymbol{m}}, \tag{5}$$

where g_F is the hyperfine Landé g-factor given by

$$g_F = g_J \frac{F(F+1) - I(I+1) + J(J+1)}{2F(F+1)}, \tag{6}$$

$\mu_B \approx 9.274 \times 10^{-24}$ J/T is the Bohr magneton constant, and $\hat{\boldsymbol{m}}$ is a unit vector describing the dipole's orientation. The factor g_J is the Landé g-factor, which for the systems we model in this project $g_J \approx 2.0023$.

In the presence of a magnetic field, our spin states experience what's called a "Zeeman shift", which changes the energy of a spin state as

$$\Delta E_{|F,m_F\rangle} = \mu_B g_F m_F B_z, \tag{7}$$

where B_z is the effective magnetic field along the z-axis. (The choice of z-axis is set by how we define our experiment).

Equation (7) gives the energy shift of one spin state. For a transition between two spin states, the shift in the transition frequency $\Delta\nu$ is

$$\Delta\nu = \frac{1}{h}\left(\Delta E_{|F_2,m_{F,2}\rangle} - \Delta E_{|F_1,m_{F,1}\rangle}\right) = \gamma_{\text{eff}} B_z, \tag{8}$$

where

$$\gamma_{\text{eff}} \equiv \frac{\mu_B}{h}\left(g_{F_2} m_{F,2} - g_{F_1} m_{F,1}\right). \tag{9}$$

Here g_{F_i} is given by Eq. (6) with $F = F_i$. Fig. 2 shows how an applied magnetic field shifts the energies of the spin states and therefore changes the frequency of light needed to drive the transition.

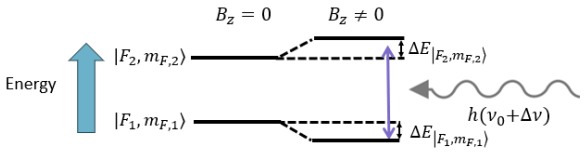


FIG. 2. Illustration of a Zeeman shift: an applied magnetic field splits and shifts the spin-state energies, changing the transition frequency.

C. Matrix-Isolated Spin Ensemble

In a full matrix-isolated spin ensemble we expect on the order of 10^{13} spins or more. Simulating how all of these spins' magnetic fields shift the energy of one atom would be very difficult. However, because the magnetic field from each spin falls off as $1/r^3$, it is reasonable to simulate only the nearest neighbors.

To do this, we create a small *test cell*, which acts as a small chunk of the full matrix. Given a test cell with side length L and dopant density ρ (the number of dopant atoms per unit volume), we expect on average

$$\bar{N} = \rho L^3 \tag{10}$$

dopant atoms in the test cell. We start by placing a single test spin at the center of the test cell. This is the spin whose transition we will study and whose frequency shift we will calculate.

To get a realistic number of neighboring spins, $N_{\text{neighbors}}$, in the cell, we draw from a Poisson distribution,

$$N_{\text{neighbors}} \sim \text{Poisson}(\bar{N}) - 1, \tag{11}$$

and subtract 1 to account for the central test spin. The $N_{\text{neighbors}}$ dopants, each indexed by i, are then placed at random positions in the cell, uniformly distributed as

$$\boldsymbol{r}_i = (x_i, y_i, z_i) \sim \text{Uniform}\left[-\frac{L}{2}, \frac{L}{2}\right]^3, \quad i = 1, \ldots, N_{\text{neighbors}}, \tag{12}$$

with the central test spin fixed at $\boldsymbol{r} = \boldsymbol{0}$. Fig. 3 shows an example of one such random test cell.

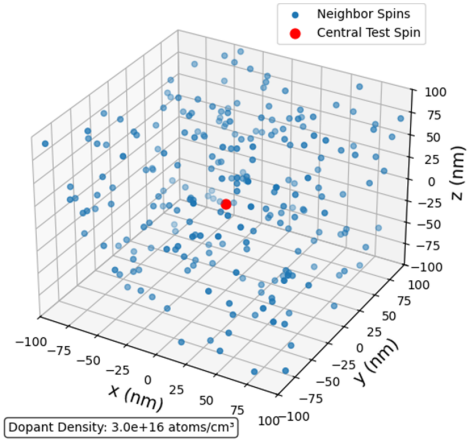


FIG. 3. Example random test cell: a central test spin (red) surrounded by $N_{\text{neighbors}}$ dopant spins (blue) at random positions.

D. Determining Spin States and Orientation

We start by choosing the spin states we wish to study in our simulation:

$$|F_1, m_{F,1}\rangle \to |F_2, m_{F,2}\rangle\,. \tag{13}$$

We define our coordinate system so that the magnetic moment of the central spin (Eq. (5)) points along the $\hat{z}$ direction.

In an ideal world, all of the neighboring spins would also point along $\hat{z}$ and be in the same initial spin state

$$|F_1, m_{F,1}\rangle\,. \tag{14}$$

In practice, experiments only achieve this alignment and state preparation with some efficiency. We model this using three probabilities:

- p_F: probability that a neighboring spin ends up in the desired hyperfine level F_1,
- p_{m_F}: probability that the spin is pumped into the desired $m_{F,1}$ state,
- p_{align}: probability that the spin's magnetic moment is aligned with the $\hat{z}$ axis.

For each neighbor spin in the test cell, the spin state is chosen as

$$F_i = \begin{cases} F_1, & \text{w.p. (with probability) } p_F, \\ F_2, & \text{w.p. } 1 - p_F, \end{cases} \tag{15}$$

$$m_{F,i} = \begin{cases} m_{F,1}, & \text{w.p. } p_{m_F}, \\ \text{Uniform}(-F_1, \ldots, +F_1), & \text{w.p. } 1 - p_{m_F}. \end{cases} \tag{16}$$

Similarly, the orientation of each spin's magnetic moment is chosen as

$$\hat{\boldsymbol{m}}_i = \begin{cases} \hat{z}, & \text{w.p. } p_{\text{align}}, \\ \text{Uniform on } S^2, & \text{w.p. } 1 - p_{\text{align}}. \end{cases} \tag{17}$$

Here S^2 is the unit sphere in three-dimensional space, meaning we pick a completely random direction.

This treatment of orientation and neighboring spin states is fairly crude. In a real experiment these variables are correlated and changing one can influence the others. However, for the order-of-magnitude estimates we are aiming for here, this simple model is sufficient.

E. Central Spin's Frequency Shift

Now that we have a test cell filled with $N_{\text{neighbors}}$ spins located at positions $\boldsymbol{r}_i$, in spin states $|F_i, m_{F,i}\rangle$, and with orientations $\hat{\boldsymbol{m}}_i$, we can calculate the magnetic field created at the central test spin.

For each neighboring spin, we use Eqs. (6), (5), and (4) to compute its magnetic moment and the resulting magnetic field $\boldsymbol{B}_i$ at the origin. We then sum these contributions and project along the z-axis to obtain the effective magnetic field at the central spin for a given test cell, indexed k:

$$B_z^{(k)} = \hat{z} \cdot \sum_{i=1}^{N_{\text{neighbors}}} \boldsymbol{B}_i. \tag{18}$$

The corresponding frequency shift for that particular test cell configuration is then

$$\Delta\nu_k = \gamma_{\text{eff}} B_z^{(k)}, \tag{19}$$

using Eq. (8).

III. ANALYSIS

The model above tells us how one specific configuration of neighboring spins shifts the transition frequency of a central test spin. In a real matrix-isolated ensemble each dopant sees a different random arrangement of neighbors. To predict what an experiment would see, we need the distribution of transition frequency shifts $\Delta\nu$ over many possible test cells.

We estimate its distribution using a Monte Carlo simulation, which here just means, repeat the random construction of test cells many times and collect statistics:

1. For a fixed choice of parameters (dopant density ρ, cell size L, and pumping / alignment efficiencies), generate one random test cell as described in Secs. II C and II D.
2. Compute the total field $B_z^{(k)}$ at the central spin from all neighboring spins and convert it into a frequency shift $\Delta\nu_k$ using Eq. (19).
3. Repeat this procedure for many independent test cells (I choose about 100,000 iterations, which produces a visually smooth distribution), producing a large sample of $\Delta\nu_k$ values.

Given this sample, we construct a histogram $H(\Delta\nu)$ by binning the $\Delta\nu_k$ values and counting how often each range of frequency shifts occurs. The histogram is our estimate of the inhomogeneous line shape, which has the inhomogeneous broadening we are about. An example histogram for one set of parameters is shown in Fig. 4.

We then fit this histogram with a Lorentzian function,

$$L(\Delta\nu) = A \frac{\Gamma/2}{(\Delta\nu)^2 + (\Gamma/2)^2}, \tag{20}$$

where A sets the overall height, and Γ is the full width at half maximum (FWHM). The FWHM is a standard way of quantifying how broad a peak is. A Lorentzian is a standard signal one may expects when doing experiments looking at the spin states of a large number of atoms.

The fitted value of Γ is interpreted as the inhomogeneous linewidth due purely to spin-spin interactions in

this model. This linewidth is directly related to the inhomogeneous coherence time,

$$T_2^* \equiv \frac{1}{\pi\Gamma}. \tag{21}$$

This coherence time is a term people will typically use to describe how "good" a signal is, where a larger T_2^* corresponds to a better signal. Since many atomic-sensor sensitivities improve when T_2^* is larger, both Γ and T_2^* are the key outputs of this simulation.

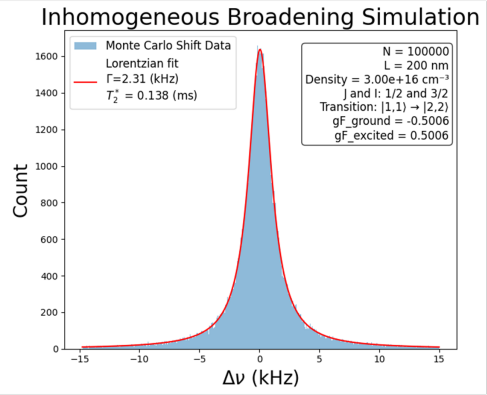


FIG. 4. Example simulated histogram of frequency shifts $\Delta\nu$ with a Lorentzian fit; the width Γ encodes the inhomogeneous broadening.

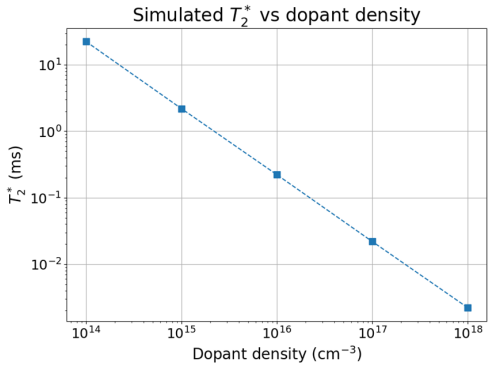


FIG. 5. Simulated inhomogeneous coherence time T_2^* as a function of dopant density ρ for a fixed Rb-85 transition. Trend is roughly $T_2^* \propto 1/\rho$.

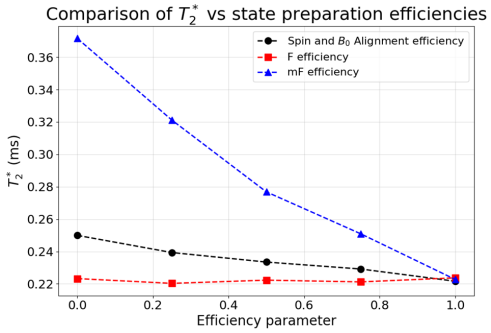


FIG. 6. Dependence of simulated T_2^* on the three efficiency parameters (p_F, p_{m_F}, p_{align}).

IV. DISCUSSION

A. General Simulation Results and Trends

First, we test how the inhomogeneous coherence time T_2^* varies when we change key simulation inputs, specifically the dopant density ρ and the neighbor spin-state efficiency parameters.

The qualitative trends in this section are similar regardless of the atom or spin-state transition we choose. For the figures shown here, we look at Rubidium-85 and the transition

$$|F_1 = 2, m_{F,1} = 2\rangle \rightarrow |F_2 = 3, m_{F,2} = 3\rangle .$$

Figure 5 shows how changing the dopant density results in different T_2^* times. We see an approximately linear relationship where the coherence follows $T_2^* \propto 1/\rho$. This makes intuitive sense as increasing the density places dopants closer together, which strengthens the spin-spin interactions and makes the frequency shifts larger, leading to a broader line and therefore a smaller T_2^*.

Figure 6 shows how varying the efficiencies described in Sec. II D changes T_2^*. In this figure, all other efficiencies are held at 1.0 (i.e. 100%) while varying each one individually. In general, changing these parameters has only a modest effect on T_2^*, especially since we care mostly about order-of-magnitude behavior rather than small factors. The only significant change we see is that decreasing the m_F efficiency can increase the coherence time by up to a factor of ~ 2 in this model.

B. Experimental Result Comparison

In this section I compare the simulation results to actual matrix-isolated spin-ensemble experiments, where inhomogeneous broadening is measured directly. In the experimental papers I use, the broadening is reported as a linewidth, which corresponds to the FWHM Γ of our Lorentzian fit, Eq. (20).

The first experimental results I look at are from Upadhyay *et al.* [3]. In their work, they study several different alkali atoms in matrix-isolated spin ensembles and report

the measured linewidths. They also vary the dopant density and identify the densities at which the linewidths begin to change noticeably. These characteristic densities are summarized in Table II. By plugging in these dopant densities and modeling the same spin-state transitions they measured, I obtain the simulated linewidths shown in Fig. 7.

Species	Dopant density ρ (atoms/cm^3)
^{39}K	5×10^{16}
^{85}Rb	$\sim 10^{18}$
^{87}Rb	$\sim 10^{18}$
^{133}Cs	10^{17}

TABLE II. Approximate dopant density concentrations used in experiments of Ref. [3].

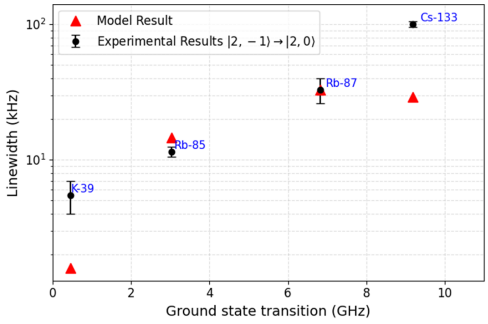


FIG. 7. Comparison of simulated linewidths in spin-spin interaction model and experimental linewidths from Upadhyay *et al.* (Fig. 10) for different alkali species at relatively high dopant densities.

From Fig. 7 we see that the model results are within the same order of magnitude of the experimental results for these high densities. Upadhyay *et al.* also perform more detailed measurements of Rubidium-85, measuring the linewidths for each possible spin-state transition of the form

$$|F = 3, m_{F,i}\rangle \rightarrow |F = 3, m_{F,i\pm1}\rangle \, .$$

Simulating these spin-state transitions, I compare the model results to the experimental results in Fig. 8. Again, the simulated linewidths and experimental linewidths agree within an order of magnitude. This agreement suggests that, at these relatively high densities, the spin-spin interaction model could explain the observed broadening.

However, Upadhyay *et al.* also report that the linewidths remain essentially unchanged when the dopant density is *decreased* by about an order of magnitude. This behavior disagrees strongly with the nearly linear density dependence predicted by Fig. 5, where spin-spin interactions alone would give narrower lines (smaller Γ or linewidth) at lower density.

Figure 9 compares the model results to experimental

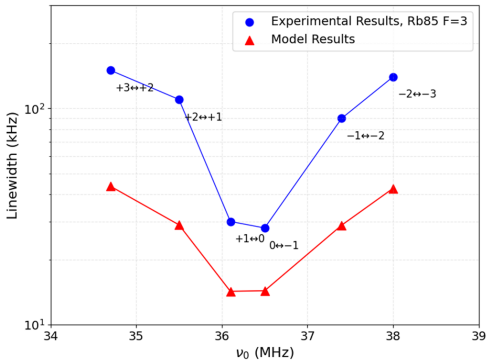


FIG. 8. Simulated vs experimental linewidths for different Rb-85 m_F transitions at high dopant density, based on Upadhyay *et al.* (Fig. 11).

measurements by Dargyte *et al.* [4]. As in Fig. 8, they study Rubidium-85 and measure the linewidths for each spin-state transition of the form

$$|F = 3, m_{F,i}\rangle \rightarrow |F = 3, m_{F,i\pm1}\rangle \, .$$

In this case, the samples are grown at a much lower dopant density, $\rho \sim 10^{16}$ atoms/cm^3, which is two orders of magnitude smaller than in the relatively high-density case presented above. Based on the model trend $T_2^* \propto 1/\rho$, if spin-spin interactions were the primary cause of inhomogeneous broadening, we would expect the linewidths at this lower density to be about two orders of magnitude smaller than in the high-density case. Instead, the experimental linewidths remain within the same order of magnitude as the high-density results.

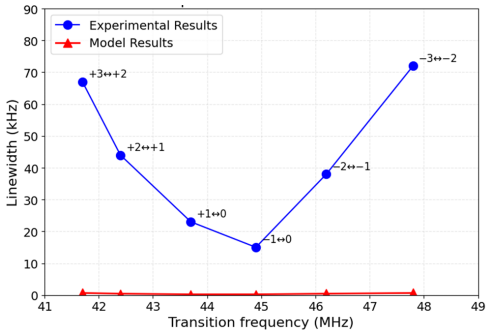


FIG. 9. Simulated vs experimental linewidths for Rb-85 m_F transitions at lower dopant density, based on Dargyte *et al.* (Fig. 12)

Looking at another experiment by Kanagin *et al.* [5], they study Sodium-23 at a density $\rho = 3 \times 10^{16}$ atoms/cm^3 and report a linewidth of 989 kHz. In contrast, using the same density and the spin-spin interaction model, I obtain a simulated linewidth of only 2.29 kHz, which is nearly three orders of magnitude smaller than the experiment.

Putting these comparisons together, the results above point to a conclusion that **spin-spin interactions alone are not the primary cause of inhomogeneous broadening at the moderate dopant densities used in most current matrix-isolated spin-ensemble experiments** ($\sim 10^{16}$ atoms/cm^3). The model predicts a strong dependence of linewidth on dopant density (Fig. 5), but the experiments show linewidths that remain roughly constant even when the density is changed by orders of magnitude (Figs. 8 and 9).

The only regime where the spin-spin model agrees reasonably well with experiment is at relatively high dopant densities, such as when Rb-85 has density $\rho \sim 10^{18}$ atoms/cm^3. Dargyte *et al.* explicitly mention that once they reach such high densities they see a much stronger relationship between density and linewidths. This suggests that at high densities spin-spin interactions could become an important, and perhaps dominant, source of inhomogeneous broadening. At the lower densities used in many current experiments, however, some other mechanism (such as structural defects) must be responsible for most of the observed broadening.

From the point of view of designing better sensors, this is actually good news. If spin-spin interactions were already the dominant broadening mechanism at current densities, then the coherence times and sensitivities would be close to a fundamental limit set by the dopant density itself. Instead, the comparison here suggests that there is still room to improve by reducing defects and other inhomogeneities in the matrix, researchers may be able to significantly decrease the linewidths and extend T_2^* before hitting the spin-spin limit predicted by this model.

V. CONCLUSION

In this project I have developed a model of spin-spin interactions between atoms in a matrix-isolated spin ensemble and used Monte Carlo methods to simulate the resulting inhomogeneous broadening. I expected this model would explain the large broadening and short T_2^* coherence times seen in experiments. However, the spin-spin interaction broadening I simulated was not large enough to explain the observed broadening in all experiments and at the dopant densities typically used in these matrix-isolated spin-ensemble systems.

My model predicts a roughly linear scaling of linewidth with density, but when I compare to data from Upadhyay *et al.*, Dargyte *et al.*, and Kanagin *et al.*, the experimental linewidths stay almost the same even when the density changes by orders of magnitude. My spin-spin interaction model only agrees well at relatively high dopant densities, where it is reasonable that spin-spin interactions start to become important and may set a fundamental limit. This points toward other effects, such as structural crystal defects, as the dominant source of inhomogeneous broadening in these systems. Regardless, this model could still be useful in providing a lower bound on inhomogeneous broadening (or upper bound on T_2^*) that one could expect with improved crystal structure.

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VI. APPENDIX

A. Experimental Data Used

All experimental data I use for my figures in Sec. IV B in their original forms is reported below.

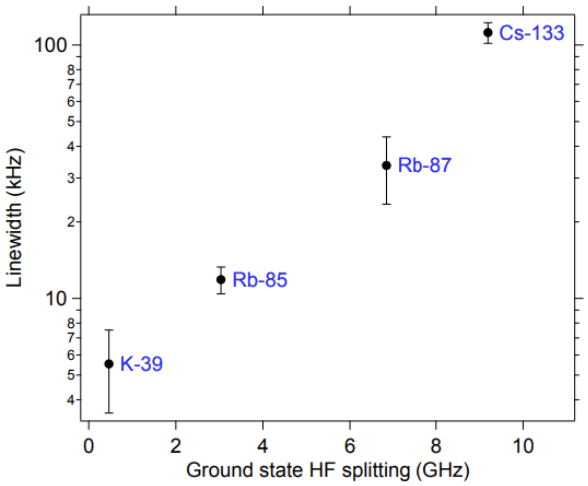


FIG. 12. T_2^*, expressed as a linewidth $= \frac{1}{\pi T_2^*}$, for the species measured. The linewidths are plotted as a function of the ground-state hyperfine splitting of each species; we believe this is the key parameter in explaining the differences in the observed linewidths, as discussed in the text. Rb and Cs data were taken at sufficiently low fields that the different Larmor precession frequencies were unresolved; the ^{39}K data was taken at similar fields but with resolved structure; the number plotted is the linewidth of the $F = 2$, $|m_F = 0\rangle$ and $|m_F = -1\rangle$ Larmor superposition. All data points were taken in the short-pulse limit. The unresolved splittings of the different Larmor precession states may be artificially broadening the ^{85}Rb linewidth at the level of $\lesssim 10\%$; less for ^{87}Rb amd ^{133}Cs.

FIG. 10. Original data and figure from [3] that was used in Fig. 7.

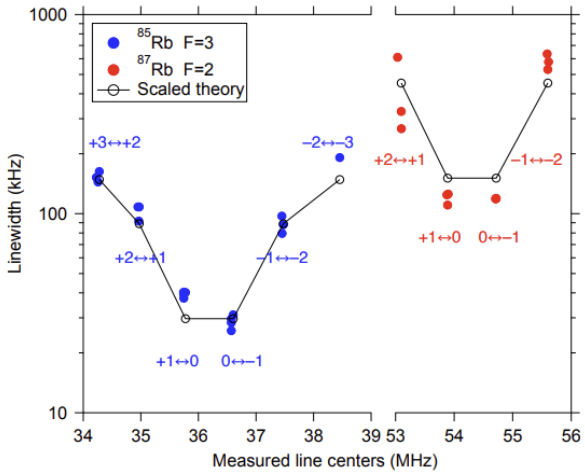


FIG. 13. The FWHM linewidths of the Rb Larmor precession superposition states at a magnetic field of 80 G, as discussed in the text. The data are labeled by the m_F states of their corresponding Larmor superposition. Plotted alongside the data is the theory of section VII A, scaled by a factor of 1.35.

FIG. 11. Original data and figure from [3] that was used in Fig. 8.

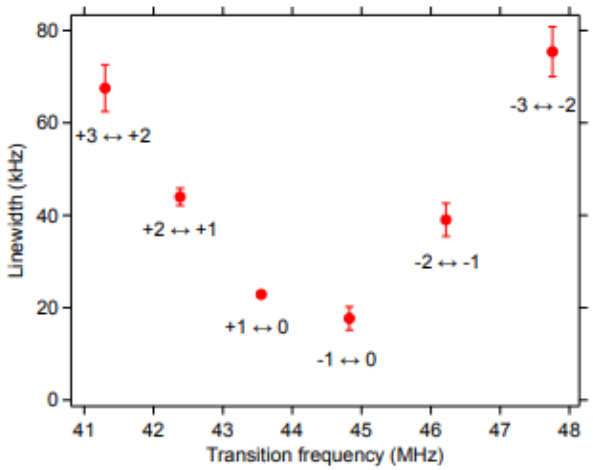


FIG. 6. The full-width-at-half-maximum (FWHM) linewidths of the transitions between adjacent m_F states in the $F = 3$ hyperfine manifold of ^{85}Rb, labeled accordingly. Taken at a magnetic field of 95 G.

FIG. 12. Original data and figure from [4] that was used in Fig. 9.

B. Python Simulation Code

The full Python code used for my spin-spin interaction induced inhomogeneous broadening simulation is show below.

Listing 1. Python code used to simulate spin-spin-induced inhomogeneous broadening

```
"""
broadening_sim.py

ES_APPM 421 Final Project:
Modeling Inhomogeneous Broadening through Spin-Spin Interactions in Matrix Isolated Spin Ensembles

Author: Isaac Cross (crossisaac114@gmail.com and/or isaaccross2026@u.northwestern.edu)
Date: December 8, 2025
"""

#Import needed packages
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
from scipy.optimize import curve_fit
from fractions import Fraction

#define constants to be used
mu0 = 4*np.pi*1e-7 #permeability of free space T m/A
muB = 9.2740100783e-24 #Bohr magneton,J/T
h = 6.62607015e-34 #planck's const J*s
gJ = 2.0023 #electron Lande g-factor, assumed free electron (alkali)

B_const = mu0/(4*np.pi) #const prefactor for B from dipole used later


#Define all helper functions to be called in Monte Carlo broadening simulation
"""
space_dopants:
Creates test cell and fills with atoms/dopants

Sec. "Matrix-Isolated Spin Ensemble"
"""
def space_dopants(density_cm3, L_box_nm):
    #get expected number of dopants
    rho_m3 = density_cm3 * 1e6 # dopant density in atoms/m^3
    V_box_m3 = (L_box_nm * 1e-9)**3 # box volume m^3
    N_mean = rho_m3 * V_box_m3 #ave num dopants in our box

    #generate number of actual dopants, assuming a Poisson distribution
    N_dopants = np.random.poisson(N_mean) - 1 #subtract 1 for the central dopant

    #assume random uniform dist in box, positions in nm, spaced from central spin @ (0,0,0)
    dopant_locations = np.random.uniform(
        low=-L_box_nm/2,
        high=L_box_nm/2,
        size=(N_dopants, 3) # Nx3 array. N dopants, at (x,y,z)
    )

    return dopant_locations

"""
lande_gF_factor
Calculates the hyperfine Lande g-factor for the spin state we are looking at

Sec. "Spin-Spin Interaction Model"
"""
```

```
58 def lande_gF_factor(J,I,F):
59     return gJ*((F*(F+1.0)-I*(I+1)+J*(J+1))/(2*F*(F+1)))
60
61
62 """
63 zeeman_coeff_calc
64 Calculatye gamma_eff
65 Sec. "Spin-Spin Interaction Model"
66 """
67 def zeeman_coeff_calc(gF_ground,mF_ground,gF_excited,mF_excited):
68     return muB*(gF_excited*mF_excited-gF_ground*mF_ground)/h #Hz/T
69
70
71
72 """
73 magnetization_dopants
74 Sec."Determing Spin States and Oreintation" in writeup
75
76 Uses gFi determined by possible Fi instead of just F as in writeup
77 as this is more efficient. Practically they are equivalent.
78 """
79 def magnetization_dopants(size,
80                           gFs, gF_pump, gF_pump_efficiency,
81                           mFs, mF_pump, mF_pump_efficiency,
82                           allignment_efficiency):
83
84     #pick mF for each state with efficiency mF_pump_efficiency to end in mF_pump
85     mF = np.empty(size, dtype=float)
86     mF_pumped_mask = np.random.rand(size)<mF_pump_efficiency
87     mF[mF_pumped_mask] = mF_pump
88     mF[~mF_pumped_mask] = np.random.choice(mFs,size=(~mF_pumped_mask).sum())
89
90     #pick gF for each state with efficiency gF_pump_efficiency to end in gF_pump
91     gF = np.empty(size, dtype=float)
92     gF_pumped_mask = np.random.rand(size)<gF_pump_efficiency
93     gF[gF_pumped_mask] = gF_pump
94     gF[~gF_pumped_mask] = np.random.choice(gFs,size=(~gF_pumped_mask).sum())
95
96     #magnitude of magnetic moment each dopant
97     m_mag = gF * mF * muB
98
99     #Magnetic moment orientation
100    m = np.tile(np.array([0., 0., 1.]), (size, 1))
101    mis = np.random.rand(size) > allignment_efficiency
102    n_mis = mis.sum()
103    if n_mis > 0:
104        #random unit vectors on S^2
105        v = np.random.normal(size=(n_mis, 3))
106        v /= np.linalg.norm(v, axis=1)[:, None]
107        m[mis] = v
108
109    #output final magnetic moment list for all dopants
110    return m_mag[:, None] * m #J/T
111
112
113 """
114 Bz_origin_calc
115
116 Calcualte magnetic field at central test spin caused by all neighboring spins
117 Sec. "Spin-Spin Interaction Model" and
118 Sec. "Central Spin's Frequency Shift"
119 """
120 def Bz_origin_calc(dopant_locations, m_dopants):
121     # Bz given ~ consts*(3rhat(m dot rhat) - m)/rmag^3
```

```
122
123     # dopant r is currently relative to origin, but want
124     # magnetic field located at -r relative to dopant
125     R = -dopant_locations*1e-9 #convert from nm to m
126
127     R_mag = np.linalg.norm(R, axis = 1)[:,None] #get magnitude of R, will be importnat for Rhat used in B
            field calc
128     R_hat = R/R_mag
129
130     m_dot_R_hat = np.sum(m_dopants*R_hat, axis=1)[:,None]
131
132     B_vecs = B_const*(3*m_dot_R_hat*R_hat - m_dopants)/(R_mag**3) #get full vector B field from each dipole
            (N,3)
133     B_z = B_vecs[:,2] #only care about Bz (N,1)
134     B_z_tot = np.sum(B_z) #sum to get total effective B field at origin
135     return B_z_tot #T
136
137
138 """
139 lorentzian
140
141 Lorentzian function for fitting
142 Sec. "ANALYSIS"
143 """
144 def lorentzian(v, A, v0, Gamma):
145     return A * (Gamma/2)/((v - v0)**2 + (Gamma/2)**2)
146
147
148 """
149 broadening_simulation
150
151 key to Monte Carlo model of spin-spin interaction and corresponding inhomogeneous broadening
152 Uses all helper functions defined above.
153 Sec. "ANALYSIS"
154 """
155 def broadening_simulation(N,density_cm3,L_box_nm,
156                           J,I,F_ground,mF_ground,
157                           F_excited,mF_excited,
158                           F_pump_efficiency,mF_pump_efficiency,allignment_efficiency,
159                           plot = True):
160
161     #get coefficients/constants
162     #gF consts
163     gF_ground = lande_gF_factor(J,I,F_ground)
164     gF_excited = lande_gF_factor(J,I,F_excited)
165     #zeem shift coeff
166     zeeman_coeff = zeeman_coeff_calc(gF_ground,mF_ground,gF_excited,mF_excited) #Hz/T
167
168     gFs =[gF_ground,gF_excited]
169     mFs = np.arange(-F_ground,F_ground+1,1)
170
171     #create empty list to fill with shifts
172     delta_freqs = np.empty(N)
173
174     #simualte new configuration and calculate shifts N times
175     for k in range(N):
176
177         dopant_locations = space_dopants(density_cm3, L_box_nm) #nm
178
179         num_dopants = dopant_locations.shape[0]
180
181         m_dopants = magnetization_dopants(num_dopants,
182                                           gFs, gF_ground, F_pump_efficiency,
183                                           mFs, mF_ground, mF_pump_efficiency,
```

```
                                    allignment_efficiency) #J/T

        magnetic_field_z = Bz_origin_calc(dopant_locations,m_dopants) #T

        delta_freqs[k] = zeeman_coeff*magnetic_field_z #Hz

    #convert from Hz to kHz
    delta_freqs_khz = delta_freqs*1e-3 #kHz

    #get guesses for p0 in lorentzian fit, and set bounds on fit

    #use central 97% of the data as the fitting range
    low_q, high_q = np.percentile(delta_freqs_khz, [3, 97])
    padding = 0.1 * (high_q - low_q)

    rangeset = (low_q - padding, high_q + padding)

    # create histogram bins but do NOT use the counts for fitting
    hist_vals, bin_edges = np.histogram(delta_freqs_khz, bins=500, range = rangeset)
    bin_centers = (bin_edges[:-1] + bin_edges[1:]) / 2

    # initial guesses
    A0 = max(hist_vals)
    x0_0 = 0

    indices = np.where(hist_vals >= A0/2)[0]
    FWHM_est = bin_centers[indices[-1]] - bin_centers[indices[0]]
    gamma0 = FWHM_est

    popt, pcov = curve_fit(lorentzian, bin_centers, hist_vals, p0=[A0, x0_0, gamma0])
    A, x0, gamma = popt

    T2_star_ms = 1/(np.pi*gamma)

    #determine if we output a plot
    if plot:
        #create figrue/plot
        fig = plt.figure(figsize=(13,6))

        #plot example dopant dist
        ax1 = fig.add_subplot(1,2,1,projection="3d")
        x_nm = dopant_locations[:,0]
        y_nm = dopant_locations[:,1]
        z_nm = dopant_locations[:,2]

        ax1.scatter(x_nm, y_nm, z_nm, s=20, label='Neighbor Spins') # neighbor spins
        ax1.scatter(0, 0, 0, s=60, color='red', label='Central Test Spin') # central test spin
        ax1.legend(loc='upper right', fontsize=10)

        ax1.set_xlabel("x (nm)", fontsize=14)
        ax1.set_ylabel("y (nm)", fontsize=14)
        ax1.set_zlabel("z (nm)",fontsize=14)
        ax1.set_xlim(-L_box_nm/2,L_box_nm/2)
        ax1.set_ylim(-L_box_nm/2,L_box_nm/2)
        ax1.set_zlim(-L_box_nm/2,L_box_nm/2)

        ax1.text(
            0.02, 0.10,0.0,
            f"Dopant Density: {density_cm3:.1e} atoms/cm",
            transform=ax1.transAxes,
            fontsize=10,
            va='top',
            ha='left',
            bbox=dict(boxstyle="round", fc="white", ec="black", alpha=0.8)
```

```
        )

        ax1.set_title("Example dopant configuration", fontsize=24)

        #Plot historgram with relevant output values, T2*, Gamma, as well as input paramters.
        ax2 = fig.add_subplot(1,2,2)

        ax2.hist(delta_freqs_khz, bins=500,range=rangeset,alpha=0.5, label="Monte Carlo Shift Data")

        x=np.linspace(rangeset[0],rangeset[1],2000)
        ax2.plot(x, lorentzian(x, *popt), 'r-',
                label=f"Lorentzian fit\n$\Gamma$={gamma:.3g} (kHz)\n"
                       f"$T_2^*$ = {T2_star_ms:.3g} (ms)")

        param_text = (
            f"N = {N}\n"
            f"L = {L_box_nm} nm\n"
            f"Density = {density_cm3:.2e} cm\n"
            f"J and I: {Fraction(J).limit_denominator()} and {Fraction(I).limit_denominator()}\n"
            f"Transition: |{F_ground},{mF_ground}  |{F_excited},{mF_excited}\n"
            f"gF_ground = {gF_ground:.4f}\n"
            f"gF_excited = {gF_excited:.4f}"
        )

        ax2.text(0.98, 0.95, param_text, fontsize=12,
                transform=plt.gca().transAxes,
                va='top', ha='right',
                bbox=dict(boxstyle="round", fc="white", ec="black", alpha=0.8))


        ax2.set_xlabel(r"$\Delta\nu$ (kHz)", fontsize=20)
        ax2.set_ylabel("Count", fontsize=20)
        ax2.legend(loc="upper left", fontsize=12)
        ax2.set_title("Inhomogeneous Broadening Simulation", fontsize=24)

        plt.tight_layout()
        plt.show()

    # return values so can run simulation and compare results in different document
    return {
        "gamma_kHz": gamma,
        "T2_star_ms": T2_star_ms,
        "delta_freqs_kHz": delta_freqs_khz,
        "F_ground": F_ground,
        "mF_ground": mF_ground,
        "F_excited": F_excited,
        "mF_excited": mF_excited
    }
```